Converging Lenses: Two Routes to a Focal Length, and a Two-Lens System

SYNOPSIS

You will find the focal length of the same lens two independent ways — once by focusing a very distant object, once from the intercept of a straight line fitted to $1/d_i$ against $1/d_o$ — and then put two lenses in series and check the magnification you measure against the one the thin lens equation predicts. The data come from the lab videos; the analysis and the report are yours.

Code LB4-M2 | **Due** See the Module 2 Lab 4 folder on Brightspace | **Pairs with** Note 6, Note 7 and Note 9 | **Where** From the videos, at home | **Report** Full lab report — use the Lab Template

The lab videos

These videos are the experiment. This lab was filmed in the physics laboratory. Every measurement comes off the video, so watch each part before you start that section and keep the handout beside you. **Take screenshots as you go** — a video-based lab gives you no apparatus to photograph, so your screenshots are the figures in your report.

Part 1 — introduction
youtube.com/watch?v=6w9ZfQI4pow

Part 2 — focal length of lens 1, introduction
youtube.com/watch?v=M6w-kiSjF00

Part 3 — focal length of lens 1, data
youtube.com/watch?v=LgihJ50Q8Po

Part 4 — focal length of lens 2, introduction
youtube.com/watch?v=-81zcwc2bYw

Part 5 — focal length of lens 2, data
youtube.com/watch?v=KJCwDv2gEZE

Part 6 — the double-lens setup, introduction
youtube.com/watch?v=ugHFunvso1s

Part 7 — the double-lens setup, data

youtube.com/watch?v=vxZ_I4Bdp70

What you need

- Optical bench
- Light source (the object)
- Image screen
- 10 cm converging lens
- 20 cm converging lens
- Vernier caliper
- The printed Ray Tracing Template, for Part III
- A straight edge

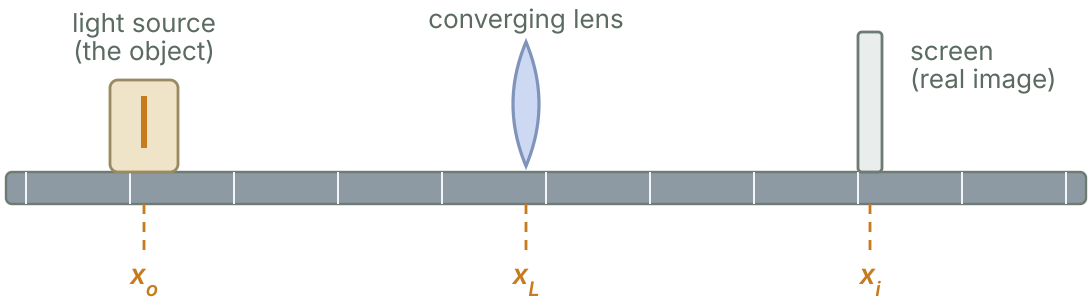


Fig. 1 The optical bench. The light source is the object; the screen is where a real image is found. Each of the three positions is read off the bench scale.

The optical bench carries the light source, one or two lenses and the screen, and every component slides along it. The bench is marked in centimetres, so what you read off it is always a *position*. Distances come later, as differences.

Part I — focal length from the theory of converging lenses

The focal length of a lens is the distance at which parallel rays come into focus.

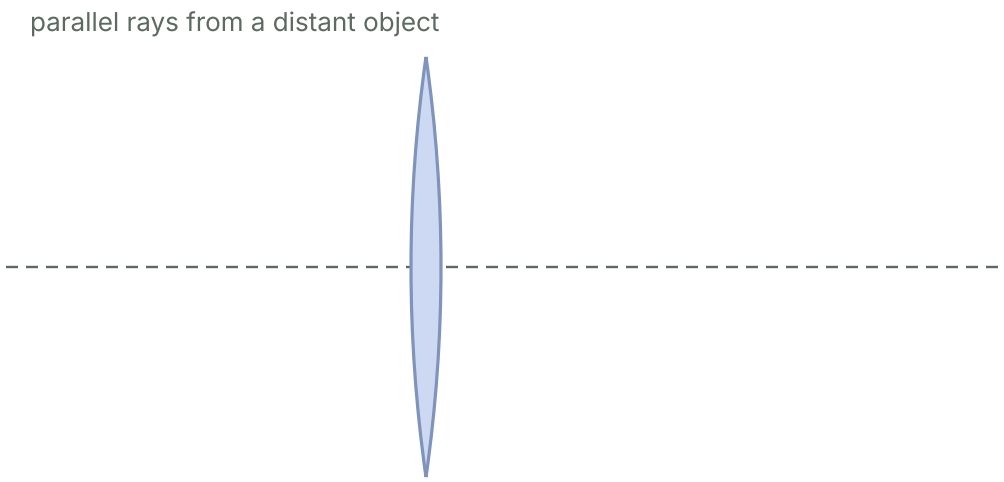


Fig. 2 *Draw the five parallel rays, show where they are brought together, label that point F and mark the focal length f.*

In practice, if an object is much farther away than the focal length, its image is in focus at the focal point — and this is the one situation where the image distance and the focal length are the same number. That gives a quick way to measure a focal length: hold the lens in one hand and a piece of paper in the other and move them until something very far away comes into focus. The gap between lens and paper is the focal length. It works best in a shadowed area with a brightly lit subject; in the video the lab is darkened and the "object" is a stand of trees well outside the window.

Procedure

1. Mount the **10 cm** converging lens on the bench at about the **140 cm** mark. Record its position as x_L in Table 3.
2. Point the optical bench at some distant object.
3. Place the screen at about the **150 cm** mark.
4. Move the screen toward or away from the lens until a sharp image forms on it. Watch the image sharpen — not the screen's position.
5. Record the screen's position as x_i in Table 3.
6. Repeat steps 1–5 twice more with the lens at different positions.
7. From the raw data, calculate the lens-to-image distance for each trial and record it in Table 4.
8. Average the three. That average is the focal length from the theory of converging lenses. Record it as f_1.
9. Calculate the percent accuracy of f_1 against the lens's stated focal length of 10 cm.

Table 3 — raw data, Part I

Trial	Lens position x_L (cm)	Image position x_i (cm)
1		
2		
3		

Table 4 — image distance, Part I

Trial	Image distance d_i (cm)
1	
2	
3	
Average	

Part II — focal length from the thin lens equation

The thin lens equation ties the focal length to the object and image distances:

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

Written that way it is already the equation of a straight line. Plot $1/d_i$ against $1/d_o$ and the line has slope −1 and an intercept of $1/f$ on the vertical axis — so it is the *intercept* that gives you the focal length.

> **Positions are not distances.** The object distance d_o is the gap between the object and the lens; the image distance d_i is the gap between the lens and the image. Both are differences of two bench readings. Carrying a position into a calculation that wants a distance is the single most common error in this lab.

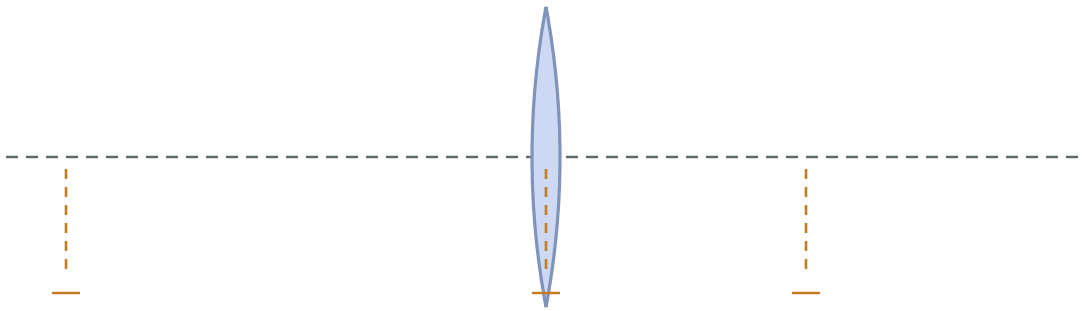


Fig. 5 *Label the three positions on the strip, draw the object and the image on the axis, then mark d_o and d_i as spans and say which two positions each one is the difference of.*

Procedure

1. Place the **10 cm** lens at the **100 cm** position and record it as x_L. Leave the lens there for the rest of Part II.
2. Place the lighted object at the **40 cm** position and record it as x_o in Table 6.
3. Place the screen beyond the lens, past the 100 cm mark.
4. Move the screen until a sharp image appears.
5. Record the screen position as x_i in Table 6.
6. Repeat steps 2–5 for x_o = 45, 50, 55, 60, 65, 70, 75, 80 and 85 cm.
7. From the raw data, calculate d_o and d_i in cm and record them in Table 7.
8. Calculate $1/d_o$ and $1/d_i$ and record them in Table 7.
9. Plot $1/d_i$ on the vertical axis against $1/d_o$ on the horizontal axis as a scatter plot in Excel.
10. Fit a linear trendline and display its equation on the chart.
11. Take the vertical-axis intercept from that equation and, using the thin lens equation, work out the focal length. Record it as f_2.
12. Calculate the percent accuracy of f_2 against the stated 10 cm.

Table 6 — raw data, Part II

Object position x_o (cm)	Image position x_i (cm)
40.0	
45.0	
50.0	
55.0	
60.0	
65.0	
70.0	
75.0	
80.0	
85.0	

Table 7 — processed data, Part II

Object distance d_o (cm)	Image distance d_i (cm)	$1/d_o$ (cm^{-1})	$1/d_i$ (cm^{-1})

Part III — the image formed by two converging lenses

With more than one lens the algebra could get heavy, but it does not have to: take the lenses one at a time and apply the thin lens equation twice. The trick is that **the image from the first lens is the object of the second**. Measure every distance from the lens in play to the object or image that belongs to it. Here you record the positions of each lens and each image, then take differences.

Procedure

1. Prepare a raw data table like Table 8.
2. Place the object at the **30 cm** mark and record it as x_o.

3. Place the **10 cm** lens at the **50 cm** mark. Call it lens 1 and record its position as x_{L1}.
4. Find the image position for this lens with the screen and record it as x_{i1}.
5. Place the **20 cm** lens at the **115 cm** position. Call it lens 2 and record its position as x_{L2}.
6. The image due to lens 1 is now the object of lens 2.
7. Move the screen beyond lens 2 and find the image it forms. Record that position as x_{i2}.
8. With the vernier caliper, measure the size of the object and of the final image. Record them as h_o and h_{i2}.
9. Determine the magnification from the heights, $M = h_{i2} / h_o$. Call it the experimental magnification 1, M_{e1}.
10. From the raw data, calculate the object distance d_{o1} (lens 1 to the object) and the image distance d_{i1} (lens 1 to the first image).
11. Calculate the magnification of the first lens, $M_1 = -d_{i1} / d_{o1}$.
12. Again from the raw data, calculate d_{o2} (lens 2 to the first image) and d_{i2} (lens 2 to the second image).
13. Calculate the magnification of the second lens, $M_2 = -d_{i2} / d_{o2}$.
14. Calculate the overall magnification $M = (M_1)(M_2)$. Call it the experimental magnification 2, M_{e2}.
15. Draw a ray diagram for this two-lens case, using the nominal focal lengths, the original lens positions x_{L1} and x_{L2}, and the object position x_o. **Use the Ray Tracing Template that comes with this lab** — print it and use a straight edge to draw the parallel ray, the focal ray and the middle ray, with the techniques from the lecture notes and the textbook.
16. Now work out the image distances and the magnifications theoretically, from the lens equation and the magnification equation. Call that magnification M_t.
17. Calculate the percent difference between M_{e1} and M_t, and between M_{e2} and M_t. Base both percentages on the theoretical magnification M_t.

Table 8 — raw data, the two-lens system

Object pos. x_o (cm)	Lens 1 pos. x_{L1} (cm)	Image 1 pos. x_{i1} (cm)	Lens 2 pos. x_{L2} (cm)	Image 2 pos. x_{i2} (cm)	Object height h_o (cm)	Image height h_{i2} (cm)

What your report must contain

A full lab report on the Lab Template. Write each of the three parts up as its own section, and give each section an experimental method, a data analysis and a conclusion of its own — then an overall conclusion at the end.

- Screenshots from the videos, one set per part, large enough to read.
- Your own sketch of each lens setup — not a copy of a figure from here.
- Tables 3, 4, 6, 7 and 8 completed, with units in the header rows.

- The $1/d_i$ against $1/d_o$ scatter plot, with its linear trendline and the equation shown on the chart.
- Both focal lengths, f_1 and f_2, each with its percent accuracy against 10 cm.
- **The Part III ray diagram, on the printed template.** Many students forget this one.
- M_{e1}, M_{e2} and M_t, and the two percent differences.
- Sample calculations — one worked instance of each distinct calculation, by hand or typed, not a spreadsheet dump.

NOTES AS YOU WORK

Questions for your discussion

1. Part I treats the image distance as the focal length. What has to be true of the object for that to hold, and roughly how far off would you be if the “distant” object were only ten focal lengths away?
2. Your two focal lengths f_1 and f_2 came from the same lens. Which do you trust more, and what in your own data supports that?
3. The fitted line should have slope −1. What does yours have, and what would a slope noticeably different from −1 tell you about the measurements?
4. Why does the intercept give the focal length while the slope does not depend on it at all?
5. In Part III you found the magnification twice, once from heights and once as $(M_1)(M_2)$. If those disagree, which measurement would you suspect first, and why?
6. The image of lens 1 is the object of lens 2 even though nothing physical sits there. Explain in a sentence or two why the second lens cannot tell the difference.

YOUR ANSWERS

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